

Spindle Type: D SERIES

DM-03

RO / LO



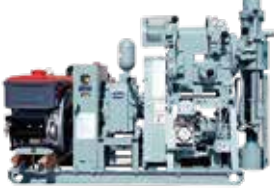
D0-D-P2-S0

RO / LO



OPTIONAL

Mounted BG-2 and Cathead Hoist



Hand Feed

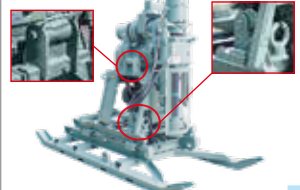


OPTIONAL

Water Pump Built in Model



Skid Base and Self-pulling System



DM-03 / D0-D

- Small size / Lightweight
- Easy to disassemble, transport, and reassemble
- Designed for safety
- Both right and left side operation available

MODEL	DM-03 (RO/LO)		D0-D-P2-S0 (RO/LO)	Water Pump Built in Model (BG-3C)
	Wire Drum Hoist	Cathead Hoist		
I.D. of Spindle	47 mm		43 mm	
Spindle Speeds	65:125 : 370 r.p.m.		(A) 60 :170 : 330 r.p.m. (B) 110 : 320 : 625 r.p.m.	
Torque (Max.)	0.54 kN-m {55 kgf-m}		(A) 0.59 kN-m {60 kgf-m} (B) 0.32 kN-m {33 kgf-m}	
Stroke of Spindle	400 (300) mm		500 (400) mm	
Thrust Force	8.4 kN {860 kgf}		17.6 kN {1790 kgf}	
Balance Force	13.8 kN {1410 kgf}		23.5 kN {2400 kgf}	
Power	3.7kW/4P or 7PS/1800 r.p.m.		3.7kW/4P or 7~9PS/1800 r.p.m.	
Dimensions (L × W × H)	1115 × 570 × 1060 mm (1115 × 570 × 960 mm)		1225×655×1285 mm (1225×655×1255) mm	1225×1110×1285 mm
Net Weight	240 (200) kg	220 (180) kg	350 (345) kg	435 kg

Spindle speed (A) is standard.

I.D. of spindle: 43mm is standard, 47mm is optional.

Drilling pump can be built in this machine.

Oil hydraulic chuck is standard equipment.

Oil hydraulic slide base is optional equipment.

USED FOR

- Geologic Survey
- Chemical Grouting
- Grouting
- Well Drilling
- Bore Piling
- Anchor Hole



D1-C-S1

RO / LO



D1-C-S1 (LO)



D1-C-S1 (RO)

Both right and left side operation available

D2-KS58

RO / LO



MODEL	D1-C-S1 (RO/LO)		D2-KS58 (RO/LO)
I.D. of Spindle	48 mm	58 mm	58 mm
Spindle Speeds	(A)65:130:240:370 r.p.m. (B)85:170:320:500 r.p.m. (C)160:310:590:900 r.p.m.	(A)50:100:185:285 r.p.m. (B)65:130:245:380 r.p.m.	(A)60:120:230:360 r.p.m. (B)100:190:360:560 r.p.m. (C)130:260:500:770 r.p.m.
Torque	(A)0.81 kN-m {83 kgf-m} (B)0.62 kN-m {63 kgf-m} (C)0.33 kN-m {34 kgf-m}	(A)1.05 kN-m {107 kgf-m} (B)0.81 kN-m {83 kgf-m}	(A)0.88 kN-m {90 kgf-m} (B)0.53 kN-m {54 kgf-m} (C)0.40 kN-m {41 kgf-m}
Stroke of Spindle	500 mm		500 mm
Thrust Force	22.1 kN {2250 kgf}		22.1 kN {2250 kgf}
Balance Force	29.5 kN {3010 kgf}		29.5 kN {3010 kgf}
Slide System	Oil Hydraulic: Stroke 300 mm		Oil Hydraulic: Stroke 400 mm
Power	5.5kW/4P or 12PS/1800 r.p.m.		5.5kW/4P or 12~15PS/1800 r.p.m.
Dimensions (L×W×H)	1580×735×1390 mm	1580×735×1430 mm	1780×870×1620 mm
Net Weight	550 kg	565 kg	680 kg

Spindle speed (A) is standard. (B)(C) is optional.

Manual chuck is standard equipment.

Spindle speed (A) is standard. (B)(C) is optional.

Manual chuck is standard equipment.

D1-C-S1

- Standard equipment
 - └ Oil hydraulic chuck
 - └ Oil hydraulic slide base
- Both right and left side operation available

D2-KS-58

- Standard equipment
 - └ Oil hydraulic chuck
 - └ Oil hydraulic slide base
- Low speed and high torque rotary spindle
- Both right and left side operation available

USED FOR

Geologic Survey

Chemical Grouting

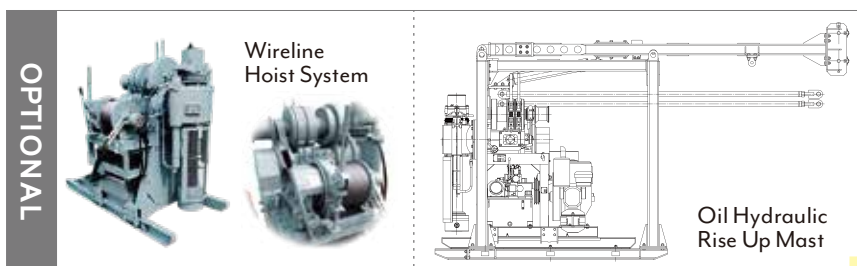
Grouting

Well Drilling

Bore Piling

Anchor Hole





D2-K92

- Standard equipment
 - └ Oil hydraulic slide base
- Maximum 90mm (HQ size) rod available
- Low speed and high torque rotary spindle
- Both right and left side operation available

D3-92C

- Standard equipment
 - └ Oil hydraulic slide base
- Long brake and hoist lever
- └ Easy hoisting operation
- Both right and left side operation available

MODEL	D2-K92-P2 (RO/LO)	D2-K92-P3 (RO/LO)	D3-92C (RO/LO)
I.D. of Spindle	92 mm		92 mm
Spindle Speeds	(A) 60:180:355:65 (R) r.p.m. (B) 100:293:577:109 (R) r.p.m.	17:50:95:18 (R) r.p.m.	113:236:452:90 (R) r.p.m.
Torque	(A) 1.18 kN-m {120 kgf-m} (B) 0.7 kN-m {73 kgf-m}	3.09 kN-m {315 kgf-m}	0.93 kN-m {95 kgf-m}
Stroke of Spindle	500 mm		500 mm
Thrust Force	22.1 kN {2260 kgf}		41.6 kN {4240 kgf}
Balance Force	29.5 kN {3010 kgf}		55.5 kN {5660 kgf}
Slide System	Oil Hydraulic: Stroke 400 mm		Oil Hydraulic: Stroke 400 mm
Power	7.5 kW/4P, 10PS/1800 r.p.m.	5.5 kW/4P, 7.5PS/1800 r.p.m.	11 kW/4P or 15PS/1800 r.p.m.
Dimensions (L x W x H)	1800 x 870 x 1620 mm	1835 x 870 x 1620 mm	1788 x 873 x 1605 mm
Net Weight	700 kg	820 kg	780 kg

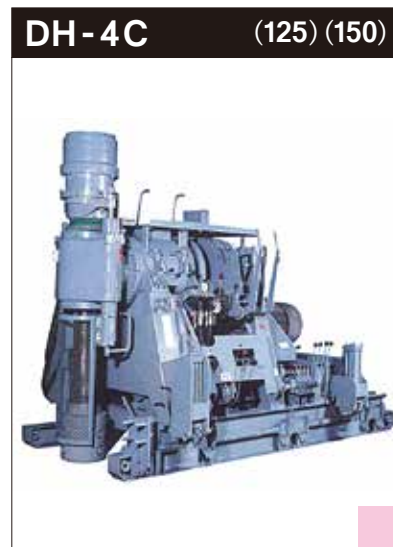
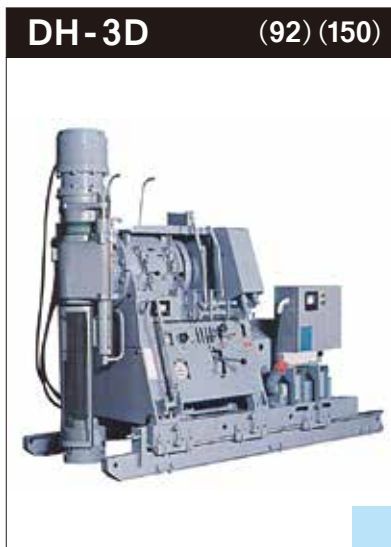
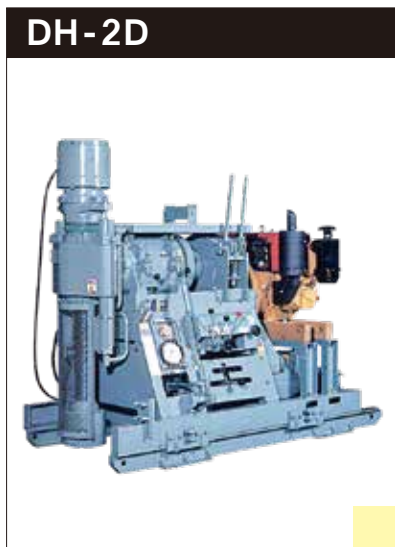
Spindle speed (A) is standard.

Oil hydraulic chuck is optional equipment.

Oil hydraulic slide base is standard equipment.

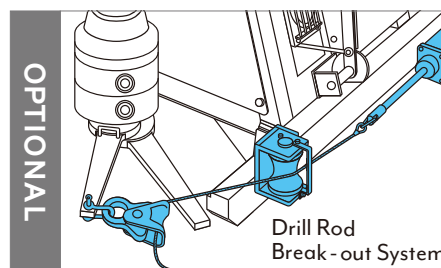
USED FOR	
Geologic Survey	
Chemical Grouting	
Grouting	
Well Drilling	
Bore Piling	
Anchor Hole	

Spindle Type: **DH** SERIES



DH-2D / DH-3D / DH-4C

- Mid- to large-sized machine with ample thrust force, balance and torque
- Used for foundation pilings, landslide prevention pilings, and water well drilling
- Standard equipment
 - └ Oil hydraulic chuck



MODEL	DH-2D	DH-3D		DH-4C	
I.D. of Spindle	79 mm	92 mm	150 mm	125 mm	150 mm
Spindle Speeds	30:105:220:26 (R) r.p.m.	30:70:200:27(R) r.p.m.	20:47:130:18(R) r.p.m.	22:50:152:18(R) r.p.m.	28:66:185:22 (R) r.p.m.
Torque	3.43 kN·m {350 kgf·m}	4.77 kN·m {490 kgf·m}	7.16 kN·m {730 kgf·m}	8.03 kN·m {820 kgf·m}	8.73 kN·m {890 kgf·m}
Stroke of Spindle	500 mm	600 mm		600 mm	
Thrust Force	29.7 kN {3030 kgf}	51.8 kN {5290 kgf}	75.4 kN {7690 kgf}	90.5 kN {9230 kgf}	
Balance Force	39.6 kN {4040 kgf}	77.0 kN {7850 kgf}	100.5 kN {10260 kgf}	120.6 kN {12310 kgf}	
Slide System	Oil Hydraulic: Stroke 500 mm	Oil Hydraulic: Stroke 500 mm		Oil Hydraulic: Stroke 600 mm	
Power	11kW/4P or 15PS/1800 r.p.m.	15kW/4P or 20PS/1800 r.p.m.		18.5kW/4P or 25PS/1800 r.p.m.	
Dimensions (L×W×H)	2075×1055×1680 mm	2530×1250×2060 mm	2670×1500×2100 mm	2920×1390×2140 mm	2985×1390×2190 mm
Net Weight	1360 kg	2140 kg	2360 kg	3170 kg	3800 kg

Jack-up system and drill rod break-out system are optional equipment.

I.D. of spindle: 125mm is standard.

Outrigger is optional equipment.



USED FOR

Geologic Survey

Bore Piling

Anchor Hole

Well Drilling

Spindle Type: **DU** SERIES

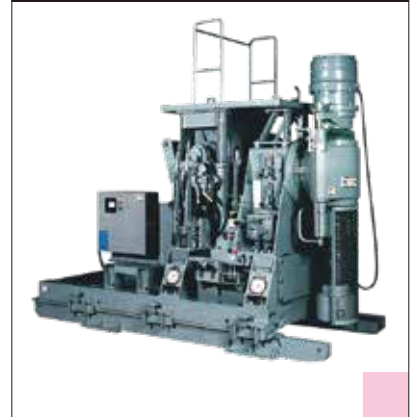
DU-4C



DU-6



DU-10

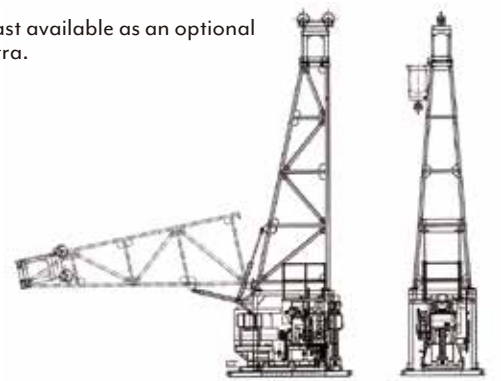


DU-4C / DU-6 / DU-10

- Twin cylinder oil hydraulic feed type
- The hexagonal drive spindle
 - └ Suitable for large diameter drilling (Exc. DU-4C)
- 6 stages of forward movement and 2 stages of reverse movement
- Oil hydraulic brakes
 - └ Easy operation and large hoisting power (Exc. DU-4C)

Most available as an optional extra.

OPTIONAL



MODEL	DU-4C	DU-6	DU-10
I.D. of Spindle	92 mm	105 mm	125 mm
Spindle Speeds	Low: 37:77:132:33(R) r.p.m. High: 220:462:792:195(R) r.p.m.	Low: 30:63:108:27(R) r.p.m. High: 180:376:646:159(R) r.p.m.	Low: 30:63:108:27(R) r.p.m. High: 180:376:646:159(R) r.p.m.
Torque	3.90 kN-m {398 kgf-m}	5.89 kN-m {601 kgf-m}	7.00 kN-m {714 kgf-m}
Stroke of Spindle	600 mm	600 mm	600 mm
Thrust Force	51.8 kN {5290 kgf}	75.4 {7690 kgf}	90.5 {9230 kgf}
Balance Force	77.0 kN {7850 kgf}	100.5 {10260 kgf}	120.6 {12310 kgf}
Slide System	Oil Hydraulic: Stroke 500 mm	Oil Hydraulic: Stroke 500 mm	Oil Hydraulic: Stroke 600 mm
Power	15kW/4P or 20PS/1800 r.p.m.	18.5 kW/4P or 25PS/1800 r.p.m.	22 kW/4P or 30PS/1800 r.p.m.
Dimensions (L x W x H)	2490 x 1089 x 1942 mm	2650 x 1170 x 2085 mm	3025 x 1330 x 2325 mm
Net Weight	1770 kg	2470 kg	3500 kg

Oil hydraulic chuck is standard equipment.

USED FOR

Geologic Survey

Well Drilling

Deep Drilling



Air Operated Drill Type: **AK** SERIES

AK-01



MODEL	AK-01
Drilling Angle	Free
Down The Hole Drill	φ 65 mm
Down The Hole Drill with Outer Casing	Drilling Diameter φ106 mm (Inside Diameter φ67 mm) Drill Rod φ50 mm, Outer Casing φ89.1 mm (φ106 mm Bit Dia.)
Spindle Speeds	0 ~ 30 r.p.m.
Torque (Max)	1.23 kN·m {125 kgf·m}
Drill Frame	Air Motor + Chain Drive
Thrust Force (Max)	7.84 kN {800 kgf}
Balance Force (Max)	7.84 kN {800 kgf}
Feed Length	1400 mm
Air Consumption	10 ~ 15 m ³ /min × 1 ~ 1.5 MPa
Dimensions (L × W × H)	2190 × 660 × 980 mm
Net Weight	Drive Unit: 230 kg
Disassembly	Drill Head: 65 kg Drill Frame: 125 kg Base: 40 kg
Operation Stand	65 kg

Feed length is 3500mm.

FEATURES

Air Drilling

Able to reduce running cost and time using air driving method.

Drill Head with Strong Air Motor

Drill head with reduction gear exerts great drilling torque.

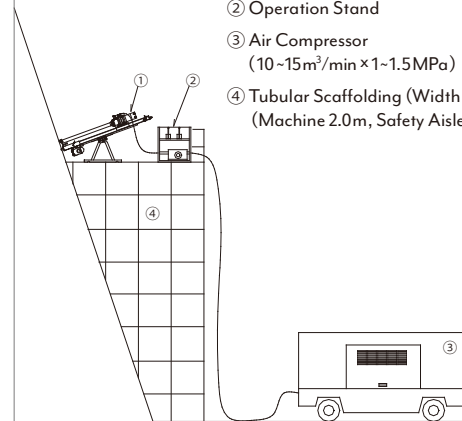
Double Tube Drilling

Makes it easy to drill into unstable stratum.

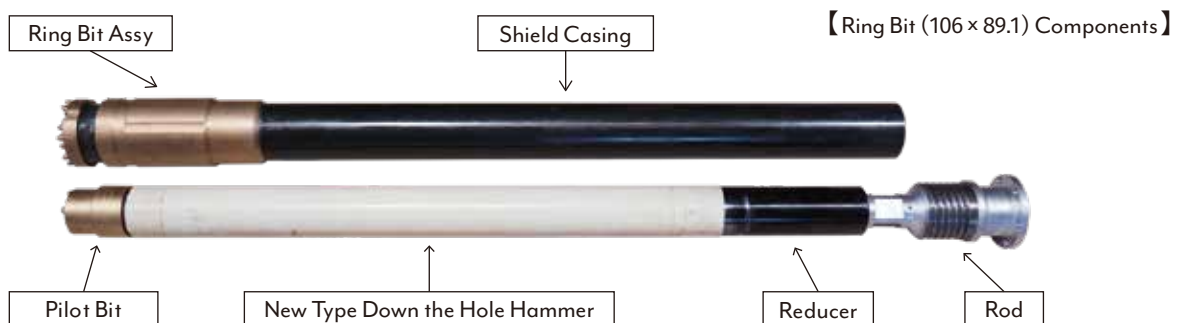
Air Kid (AK-01) LAYOUT

Machine and Tubular Scaffolding

- ① AK-01 Body
- ② Operation Stand
- ③ Air Compressor (10~15m³/min × 1~1.5MPa)
- ④ Tubular Scaffolding (Width 2.5m) (Machine 2.0m, Safety Aisle 0.5m)



Double Tube Type System



SETTING



Single Tube Type Method (Steep Slope Failure Prevention)



Double Tube Type Method (Rock Bolt)



USED FOR

Rock Bolt

Top Drive Type: **K** SERIES

K0-C



K1-B



MODEL	K0-C	K1-B
Drilling Angle	0~90°	0~48°
Diameter of Usable Pipe	Inner Pipe : 76.3 mm(Max) Outer Pipe: 139.8 mm(Max)	Inner Pipe : 89.1 mm(Max) Outer Pipe: 165 mm(Max)
Spindle Speeds	High Speed: 50Hz: 0~34 r.p.m. 60Hz: 0~40 r.p.m. Low Speed : 50Hz: 0~17 r.p.m. 60Hz: 0~20 r.p.m.	High Speed: 50Hz: 0~50 r.p.m. 60Hz: 0~60 r.p.m. Low Speed : 50Hz: 0~25 r.p.m. 60Hz: 0~30 r.p.m.
Torque (Max.)	High Speed: 2.15 kN·m {219 kgf·m} Low Speed : 4.3 kN·m {438 kgf·m}	High Speed: 3 kN·m {306 kgf·m} Low Speed : 6 kN·m {612 kgf·m}
Feed Speeds	12 m/min (Max)	14 m/min (Max)
Feed Length	1400 mm	1400 mm
Spindle Stroke	150 mm	250 mm
Thrust Force	Inner Pipe : 12 kN {1224kgf} Outer Pipe: 25 kN {2550kgf}	Inner Pipe : 35 kN {3570kgf} Outer Pipe: 65 kN {6630kgf}
Balance Force	Inner Pipe : 17.5 kN {1785kgf} Outer Pipe: 25 kN {2550kgf}	Inner Pipe : 55 kN {5610kgf} Outer Pipe: 65 kN {6630kgf}
Power Unit	Unit: 15kW/4PS Oil Heater: 3kW Oil Cooler: 0.1kW	Oil Hydraulic Unit: 30kW/4P
Dimensions (L x W x H)	Machine: 2620 x 870 x 1130 mm Oil Hydraulic Unit: 1300 x 610 x 1250 mm	Machine: 2720 x 1350 x 1400 mm Oil Hydraulic Unit: 1700 x 750 x 1155 mm
Net Weight	Machine: 830 kg Oil Hyd.Unit: 650 kg	Machine: 1440 kg Oil Hyd.Unit: 1000 kg

Other Type: Stroke of thrust head is 1900mm and 2400mm.

Oil hydraulic chuck is optional equipment.

K-0C / K1-B

- Suitable for water drainage against landslide / Anchor hole for retaining wall
- All oil hydraulic operation
- Double tube drilling
- Low speed and high torque rotary spindle

FEATURES

1

Double Tube Drilling Method

Makes it easy to drill into unstable stratum.

2

Original Drilling Method

Can increase drilling speed by using Double Tube Drilling System and Power Ring Bit System.

3

Speed Switching Valve

Makes it increase excavation efficiency.

4

Base For Incline Drilling Style

Smooth to adjust drilling angle.

Power Ring Bit System (Japanese Patent)



Rotary percussion drilling
Excellent in excavation efficiency

Speed Switching Valve K0-C



Easy to switch speed

Oil Hydraulic Holder K0-C



Chucking firmly

High Speed Inner Head



Diamond bit available

Base for Vertical Drilling Style K0-C



Wide range of drilling angle

Core Sampling



Core sampled by K1-B with high speed inner head

USED FOR

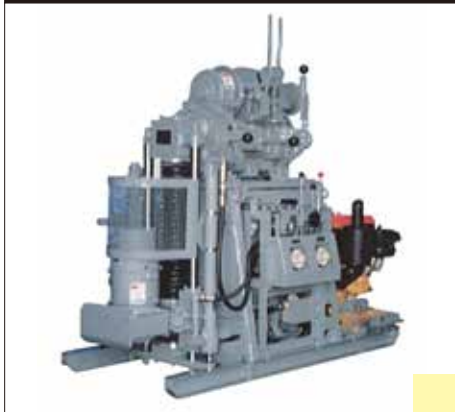
Water Drainage

Anchor Hole



Low Head Type: **LH/DW** SERIES

D0-D LH-S0 RO / LO



MODEL	D0-D LH-S0 (RO/LO)	D2-K-P3-LH92
I.D. of Spindle	43 mm	92 mm
Spindle Speeds	(A) 60:170:330 r.p.m. (B) 110:320:625 r.p.m.	14:40:80:15(R) r.p.m.
Torque	(A) 0.59 kN·m {60 kgf·m} (B) 0.32 kN·m {33 kgf·m}	3.90 kN·m {398 kgf·m}
Stroke of Spindle	500 mm	500 mm
Thrust Force	10.7 kN {1090 kgf}	22.1 kN {2250 kgf}
Balance Force	17.6 kN {1800 kgf}	29.5 kN {3010 kgf}
Power	3.7kW/4P or 5~7.5PS/1800 r.p.m. (5.5kW/4P, Including Pump)	5.5kW/4P or 7.5PS/1800 r.p.m.
Dimensions (L × W × H)	1420 × 675 × 1415 mm (1420 × 895 × 1415 mm)	2075 × 870 × 1685 mm
Net Weight	460 kg	950 kg

Spindle speed (A) is standard.

Oil hydraulic chuck is optional equipment.

Oil hydraulic slide base is standard equipment.

D2-K-P3-LH92



LH Series

- Drilling machine with the swivel head set up in a low position
- Quick and safe attachment/detachment of rods
- Suitable for chemical grouting
- Standard equipment
 - └ Oil hydraulic chuck

DW-2



MODEL	DW-2	DW-3
I.D. of Spindle	92 mm	92 mm
Spindle Speeds	30:70:220:26(R) r.p.m.	20:70:145:18(R) r.p.m.
Torque	3.50 kN·m {357 kgf·m}	5.25 kN·m {535 kgf·m}
Stroke of Spindle	600 mm	1000 mm
Thrust Force	36.3 kN {3700 kgf}	51.8 kN {5290 kgf}
Balance Force	53.9 kN {5500 kgf}	77.0 kN {7850 kgf}
Slide System	Oil Hydraulic: Stroke 500 mm	Oil Hydraulic: Stroke 500 mm
Power	11kW/4P or 15PS/1800 r.p.m.	11kW/4P or 15PS/1800 r.p.m.
Dimensions (L × W × H)	2210 × 1020 × 1525 mm	2270 × 1205 × 2145 mm
Net Weight	1850 kg	1980 kg

DW-3



DW-2 / DW-3

- Centralized operation levers
 - └ Easy operation
- Box-shape framework
- Parallel-winding drum
- Standard equipment
 - └ Oil hydraulic chuck

For Soil Improvement: **Chemical Grouting** METHOD

D2-JS-P3



D2-JS-P3-LH



CPW-5D



MODEL	D2-JS-P3	D2-JS-P3-LH
I.D. of Spindle	48 mm	48 mm
Spindle Speeds	55 : 155 : 310 r.p.m.	40 : 115 : 230 r.p.m.
Torque	0.95 kN-m {97 kgf-m}	1.31 kN-m {134 kgf-m}
Stroke of Spindle	500 mm	500 mm
Thrust Force	22.1 kN {2250 kgf}	22.1 kN {2250 kgf}
Balance Force	29.5 kN {3010 kgf}	29.5 kN {3010 kgf}
Power	5.5kW/4P or 7.5PS/1800 r.p.m.	5.5kW/4P or 7.5PS/1800 r.p.m.
Dimensions (L x W x H)	1430 x 800 x 1390 mm	1660 x 800 x 1320 mm
Net Weight	470 kg	600 kg

Manual chuck is optional equipment.

MODEL	CPW-5D
Type	Two Separately Ports, Four Ceramic Plungers, Single Acting
No. of Plunger	4
Discharge Volume	2~20 x 2 L/min
Discharge Pressure	3.4 MPa {35 kgf-cm ² }
Plunger Diameter	40 mm
Plunger Stroke	45 mm
Power	3.7kW/6P
Dimensions (L x W x H)	1225 x 575 x 800 mm
Net Weight	300 kg

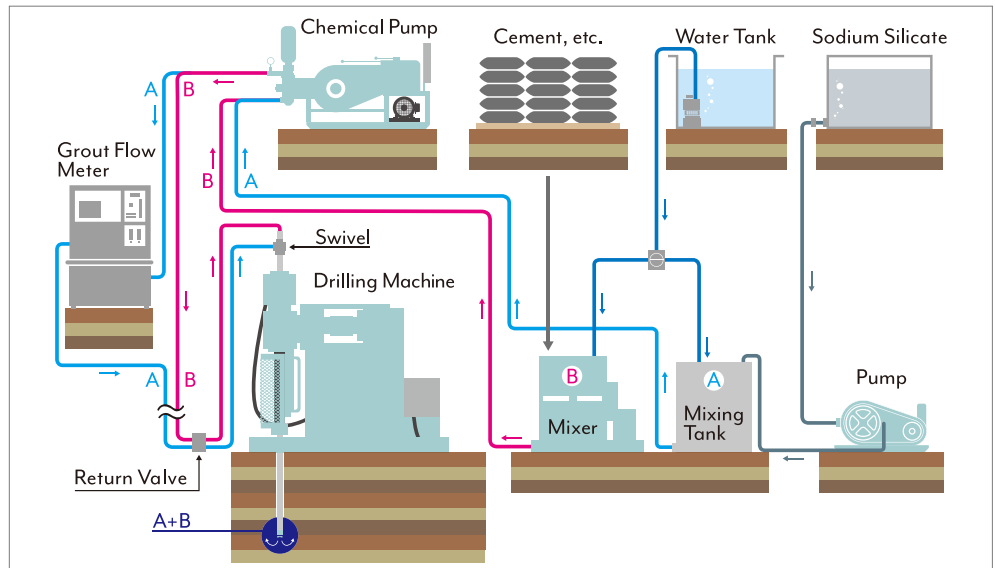
Net Weight: Setting with electric motor and control box.

D2-JS-P3

- Standard equipment
 - └ Oil hydraulic chuck
- Low speed and high torque rotary spindle

D2-JS-P3-LH

- Drilling machine with the swivel head set up in a low position
- Quick and safe attachment/detachment of rods
- Standard equipment
 - └ Oil hydraulic chuck



USED FOR

Chemical Grouting

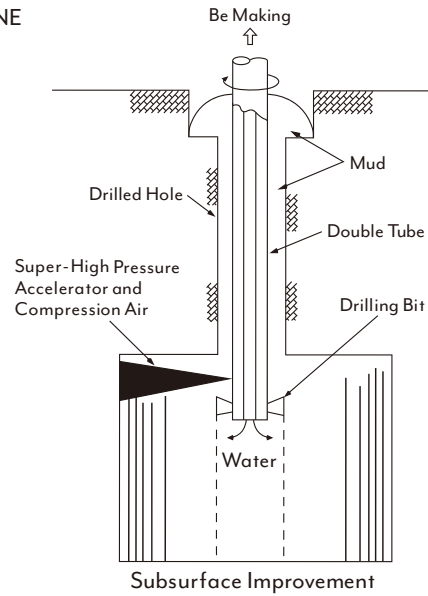


D2-K92-P2-JSG

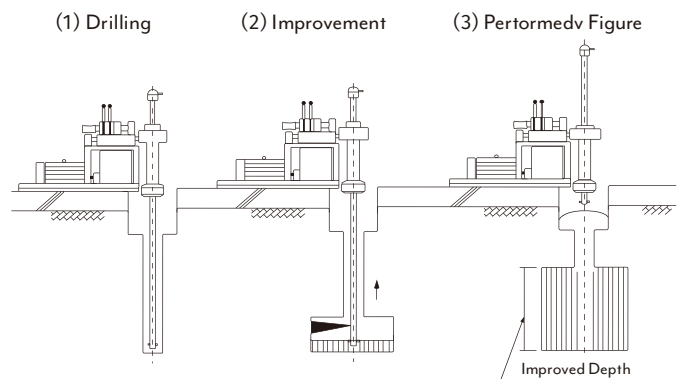


MODEL	D2-K92-P2-JSG
I.D. of Spindle	92 mm
Spindle Speeds	1st : 6~60 r.p.m. 2nd: 22~224 r.p.m. 3rd: 34~346 r.p.m. (R): 5~50 r.p.m.
Torque	1.20 kN·m {122 kgf·m}
Stroke of Spindle	500 mm
Auto Step Up	Step interval (Min): 25 mm
System	Wait Time: 0.1~999 sec
Thrust Force	20.2 kN {2060 kgf}
Balance Force	27.0 kN {2750 kgf}
Type of Hoist	Electric Hoist
Hoisting Force	1.0 kN {100 kgf}
Rope Speed	14 m/min
Slide System	Oil Hydraulic: stroke 400 mm
Power	For Spindle: 7.5 kW/4P Inverted Motor For Oil Pump: 2.2 kW/4P For Hoist: 0.4 kW/4P, 1/45 Gear Motor
Dimensions (L×W×H)	1900×1200×1650 mm
Net Weight	1000 kg

OUT LINE



Soil improvement with JSG (Jumbo Jet Special Grout) METHOD



Oil hydraulic chuck and Jack-up system are standard equipment.

